

OWUS

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Chapters

Overview of OWUS

Chapter 1: Ecohydrological Data Preparation

Chapter 2: Stochastic Soil Water Balance Model (SSWM)

Chapter 3: Water Use Strategy Framework

Chapter 4: Model Parameter Inference (MCMC)

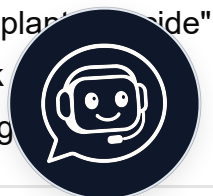
Chapter 5: Scientific Visualization

Show Raw Markdown

Chapter 3: Water Use Strategy Framework

Welcome back! In [Chapter 1: Ecohydrological Data Preparation](#), we learned how to get all our environmental ingredients ready. Then, in [Chapter 2: Stochastic Soil Water Balance Model \(SSWM\)](#), we used these ingredients with our SSWM "chef" to understand a plant's long-term water budget.

Now, we're going to use what we've learned in an exciting way: by exploring how plants "decide" to use water. This chapter introduces the **Water Use Strategy Framework**. Think of it as a special tool that lets us compare different "game plans" plants might use to manage



What Problem Does the Water Use Strategy Framework Solve? Comparing Plant "Game Plans"!

Imagine you're a coach trying to understand how different sports teams play. Some coaches might base their strategy on **historical performance**: "When it rains, we always run the ball more." This is like looking at past data to inform decisions. Other coaches might use a **theoretically "best" approach**: "To maximize our chances of winning, we *should* pass more in this situation." This is about optimizing for a desired outcome.

Plants, too, have different "game plans" for managing water:

- **Drought-Avoider**: A plant that tries to use water conservatively, closing its stomata (tiny pores on leaves) early to avoid severe stress, even if it means less growth.
- **Drought-Tolerator**: A plant that keeps its stomata open longer, trying to grow even under stress, perhaps having more robust water transport systems.

The central problem: We want to understand and compare these different plant "game plans" or **water use strategies**. Do plants actually follow strategies that are "best" for them, or do their strategies simply reflect what has worked historically in their environment? The **OWUS** project's "Water Use Strategy Framework" provides the conceptual and computational tools to make this comparison.

A concrete example: For our 'AT-Neu' grassland site, we might ask:

1. What water use strategy does the grass *actually* show, based on our observed soil moisture data? (This is like the "historical performance" game plan).
2. What would be the *optimal* water use strategy for this grass to maximize its long-term water use efficiency or minimize stress? (This is like the "theoretically best" game plan).
3. How do these two strategies compare in terms of actual water use, stress levels, and soil moisture patterns?

This framework helps us set up these comparisons to gain deeper insights into how plants thrive in their environment.

Key Concepts: Understanding Different Strategies

The framework allows us to classify and compare strategies based on how plant parameters (like how sensitive their stomata are to water stress, or how well their internal plumbing (xylem) transports water) are set.

1. Plant Water Use Strategies: The "Game Plans"

These are the characteristic ways plants regulate their water uptake and loss in response to varying soil moisture. They are mainly governed by specific plant parameters that determine their hydraulic functioning and stomatal control. For example, a plant's "strategy" defines when its stomata will close in response to drying soil.

2. Data-Driven Strategies: Based on "Past Performance"

In a "data-driven" strategy, we figure out a plant's parameters (its "game plan" settings) by **fitting our SSWM (Chapter 2) to match observed soil moisture patterns** from our [Ecohydrological Data Preparation \(Chapter 1\)](#).

- **Analogy:** You're analyzing a basketball team. You look at their past game statistics (observational data) and notice that when their star player is on the court, they pass 60% of the time. So, your "data-driven" strategy for that player's passing percentage would be 60%. It reflects *what happened*.

3. Optimality-Based Strategies: Aiming for the "Best Outcome"

In an "optimality-based" strategy, we figure out a plant's parameters by **optimizing them for specific performance metrics** from the SSWM (like maximizing long-term transpiration or water use efficiency, or minimizing water stress).

- **Analogy:** For the same basketball player, you calculate that to achieve the *highest possible score*, they *should* pass 75% of the time, even if they historically passed less. This is about what's "best" for a defined goal.

4. The Framework's Role: The Comparison Tool

The "Water Use Strategy Framework" isn't a new model, but rather a **conceptual structure and computational workflow** that allows us to:

1. **Define** different parameter sets representing various strategies (e.g., one set from data-fitting, another from optimization).
2. **Input** these parameter sets into the [Stochastic Soil Water Balance Model \(SSWM\)](#).
3. **Compare** the SSWM's outputs (like soil moisture distribution, mean transpiration, mean stress, water use efficiency `epsilon`) for each strategy. This comparison helps us understand the ecological implications of each "game plan."

How to Use the Framework: A Conceptual Comparison

The core of using this framework is to feed the [Stochastic Soil Water Balance Model \(SSWM\)](#) with different sets of plant parameters, each representing a distinct water use strategy. While the *derivation* of these specific parameters is a more advanced topic (covered in [Chapter 4: Model](#)

Parameter Inference (MCMC)), we can conceptually demonstrate how to use SSWM to *evaluate* pre-defined strategies.

Let's assume, for our 'AT-Neu' grassland, we've already obtained two sets of plant parameters:

- `params_data_driven`: Derived by fitting SSWM to observed soil moisture (data-driven strategy).
- `params_optimality`: Derived by optimizing SSWM's water use efficiency (`epsilon`) (optimality-based strategy).

Now, we'll use the SSWM to see what outcomes these two strategies predict:

```
1 import pandas as pd
2 import pickle
3 from sswm import SM_C_H # Our SSWM model
4
5 site_id = 'AT-Neu'
6
7 # --- 1. Load the pre-defined parameter sets for different strategies ---
8 # In a real workflow, these would come from the output of Chapter 4.
9 # For this chapter, imagine they are already available.
10
11 # Load base parameters for the site (from Chapter 1)
12 base_params_filepath = f'DATA/WUS/WUS_input_data/{site_id}_parameters.pickle'
13 with open(base_params_filepath, 'rb') as f:
14     base_site_params = pickle.load(f)
15
16 # Create example 'data-driven' parameters (simplified for demo)
17 # Imagine 'Pg50' (stomatal sensitivity) was inferred to be -1.5 MPa
18 # and 'k_xl_max' (xylem conductivity) was inferred to be 0.0008 m/day/MPa
19 params_data_driven_dict = base_site_params.copy()
20 params_data_driven_dict['Pg50'] = -1.5
21 params_data_driven_dict['k_xl_max'] = 0.0008
22
23 # Create example 'optimality-based' parameters (simplified for demo)
24 # Imagine optimization found 'Pg50' = -1.0 MPa (less sensitive)
25 # and 'k_xl_max' = 0.0012 m/day/MPa (higher conductivity)
26 params_optimality_dict = base_site_params.copy()
27 params_optimality_dict['Pg50'] = -1.0
28 params_optimality_dict['k_xl_max'] = 0.0012
29
30 # Convert dictionaries to the ordered list format expected by SM_C_H
```

```

31 def prepare_sswm_params(param_dict):
32     # This function would map dict keys to the specific ordered list
33     # The actual order and full list is defined in sswm.py
34     # This is a simplified representation.
35     return [
36         param_dict['Td'], param_dict['rf_alpha'], param_dict['rf_lambda'],
37         param_dict['ETo'], param_dict['LAI'], param_dict['RAI'],
38         param_dict['hc'], param_dict['Zm'], param_dict['Ps0'],
39         param_dict['b'], param_dict['Ks'], param_dict['n'],
40         param_dict['s_h'], param_dict['s_fc'], param_dict['k_xl_max'],
41         param_dict['Px50'], param_dict['Pg50'], param_dict['T_GS'],
42     ]
43
44 sswm_params_dd = prepare_sswm_params(params_data_driven_dict)
45 sswm_params_opt = prepare_sswm_params(params_optimality_dict)
46
47 # --- 2. Initialize SSWM for each strategy and get results ---
48
49 print(f"--- Evaluating Data-Driven Strategy for {site_id} ---")
50 sswm_dd = SM_C_H(sswm_params_dd)
51 if not np.isnan(sswm_dd.epsilon):
52     print(f"  Mean Transpiration (mm/day): {sswm_dd.get_mean_t(sswm_dd.p0):.2f}")
53     print(f"  Water Use Efficiency (epsilon): {sswm_dd.epsilon:.2f}")
54     print(f"  Mean Stress: {sswm_dd.get_mean_stress(sswm_dd.p0):.2f}")
55 else:
56     print("  Data-driven strategy could not be evaluated.")
57
58 print(f"\n--- Evaluating Optimality-Based Strategy for {site_id} ---")
59 sswm_opt = SM_C_H(sswm_params_opt)
60 if not np.isnan(sswm_opt.epsilon):
61     print(f"  Mean Transpiration (mm/day): {sswm_opt.get_mean_t(sswm_opt.p0):.2f}")
62     print(f"  Water Use Efficiency (epsilon): {sswm_opt.epsilon:.2f}")
63     print(f"  Mean Stress: {sswm_opt.get_mean_stress(sswm_opt.p0):.2f}")
64 else:
65     print("  Optimality-based strategy could not be evaluated.")
66
67 print("\n--- Comparison Complete ---")
68

```

What happens here?

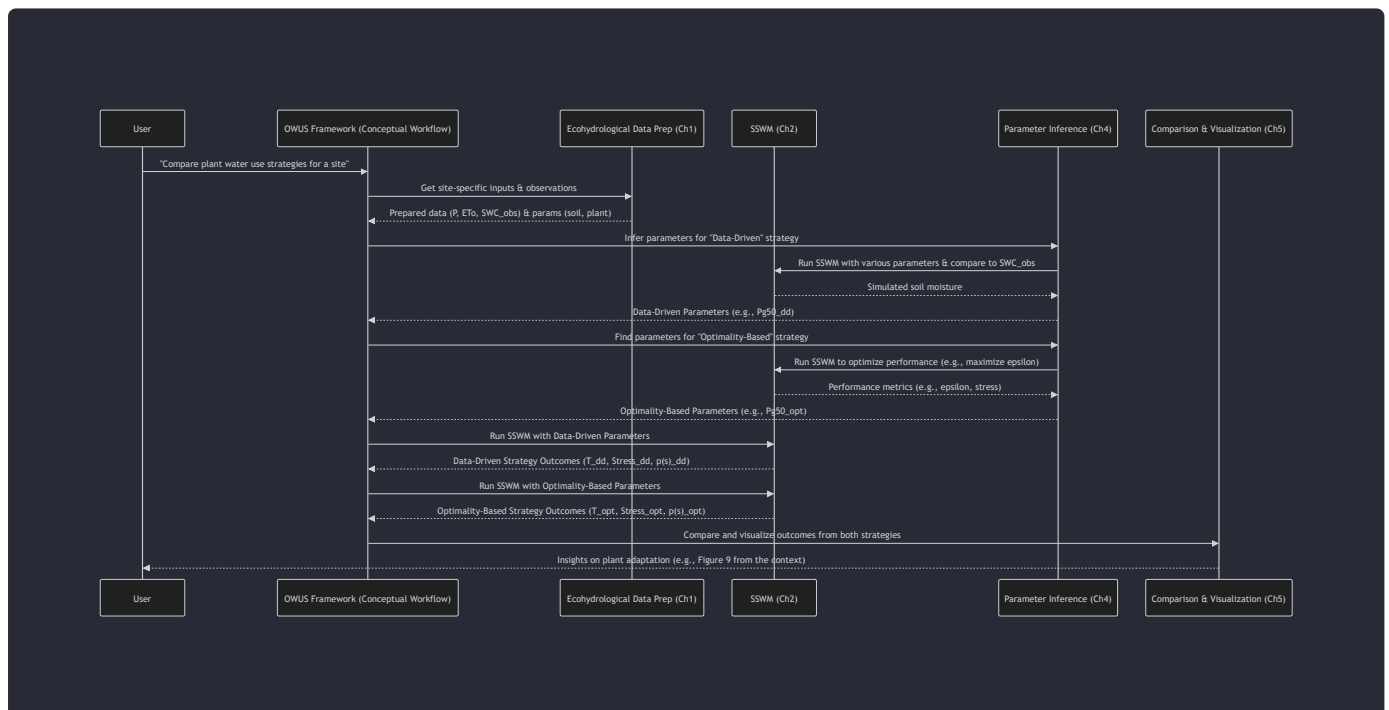
- We load two hypothetical sets of parameters (`sswm_params_dd` and `sswm_params_opt`), each representing a different plant water use strategy.
- We then *initialize* the Stochastic Soil Water Balance Model (SSWM) twice, once for each strategy's parameters.
- After initialization, the SSWM automatically calculates the long-term water balance for that specific strategy.
- We then print out key outputs for each, such as mean transpiration and water use efficiency (`epsilon`). This allows us to **compare the predicted outcomes** of the different strategies.

This direct comparison of outputs is the essence of the Water Use Strategy Framework. It helps us answer questions like: "If the plant played optimally, would its transpiration be higher or lower than what our data suggests?"

Under the Hood: The Framework as a Workflow

The Water Use Strategy Framework itself isn't a single Python class or function; it's a **workflow** that combines the tools from previous and upcoming chapters. It's about setting up a structured comparison.

Here's a simplified conceptual view of how this framework (as a workflow) operates:



Key Takeaways for "Under the Hood":

- **Leveraging Existing Tools:** The framework doesn't introduce entirely new computational models, but rather a structured way to *use* the Stochastic Soil Water Balance Model (SSWM) with different sets of parameters.

- **Parameter Derivation:** The crucial part is how we obtain the parameters for "data-driven" vs. "optimality-based" strategies. This involves either statistical inference (fitting to observed data) or mathematical optimization (finding parameters that yield the "best" performance). These advanced methods are explored in the next chapter.
- **Comparative Analysis:** The framework's value lies in its ability to directly compare the outputs of SSWM under these different strategic assumptions, providing a robust way to analyze plant water use behavior.

Conclusion

In this chapter, we've introduced the **Water Use Strategy Framework** as a crucial tool for understanding how plants manage water. We learned that it allows us to compare different "game plans": "data-driven" strategies, which reflect observed behavior, and "optimality-based" strategies, which aim for the best possible outcome. By conceptually running our Stochastic Soil Water Balance Model (SSWM) with parameters representing these strategies, we can compare their predicted ecohydrological outcomes.

This framework sets the stage for a deeper investigation into how we actually *find* these strategic parameters.

Are you ready to learn how `OWUS` infers the specific parameters that define these different plant water use strategies? Then let's move on to the next chapter!

[Next Chapter: Model Parameter Inference \(MCMC\)](#)

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